



**MASTERMIND
INDIA**

Spirit of India Special

Dharamveer KAMBOJ, 45

INNOVATION

**Multi-purpose food
processing machine**

COST

Rs 1.2 lakh

WHAT IT DOES

**Extracts juices and
also serves as a
pressure cooker**

| By Nivedita Mukherjee |

Innovation in India is not happening in shiny chrome-and-steel institutions alone but in bucolic backyards that no one pays attention to. Backyards such as the Damla village in Yamuna Nagar, Haryana, where farmers like Dharamveer Kamboj, 45, are redefining enterprise with a multi-purpose food processing machine aimed at providing a one-stop solution for farm processing of herbs, flowers and fruits. The device, costing Rs 1.2 lakh and fabricated in his village, will help cut down the colossal wastage in India's agricultural supply chain.

While Kamboj's links with land and an inherent curiosity about

nature owe much to his father who operated a small flour mill and jaggery processing plant, it was actually his move to Delhi to work as a rickshaw-puller that sowed the seeds of the innovation. The traders he ferried to the Khari Baoli area of Old Delhi paid hefty amounts for herbs and an intrigued Kamboj soon gathered information about the income they could fetch.

An accident in 1987 forced him to return to Yamuna Nagar and he joined a training programme in improving agricultural practices and organic farming techniques. In 1990, even as he became the first farmer in his area to cultivate hybrid tomatoes, the creative streak resulted in the development of devices like a battery-operated spraying machine using a tape-

recorder motor, an insect trapping device and a farm implement customised for ploughing without disturbing the sugarcane crop. Yet, this was not what he wanted and after some more trials, he built another device by fitting an old auto engine with components that could be used for processing aloe vera and amla. This machine, costing less than Rs 8,000, became so popular that people from nearby places started coming in to catch a demonstration.

In 2003, he started work on the final design of the machine and in March 2005 had the first prototype ready. The first machine developed a problem of overheating when the material to be processed was less. He tried to improve but could not elimi-

